

Multiple Choice

1. **Answer:** D

Options: A) 23.56 GHz; B) 50 MHz; C) 100 MHz; D) 42.58 MHz; E) 1 GHz

Note: Correct option: D - 42.58 MHz

2. **Answer:** A

Options: A) 0.5 N; B) 0.2 N; C) 0.75 N; D) 0.1 N; E) 1 N

Note: Correct option: A - 0.5 N

3. **Answer:** B

Options: A) 4 sigma and 1 pi; B) 3 sigma and 1 pi; C) 0 sigma and 4 pi; D) 3 sigma and 2 pi; E) 2 sigma and 1 pi

Note: Correct option: B - 3 sigma and 1 pi

4. **Answer:** C

Options: A) 9.30 g O; B) 2.50 g O; C) 4.00 g O; D) 5.00 g O; E) 3.10 g O

Note: Correct option: C - 4.00 g O

5. **Answer:** A

Options: A) $4.20 \times 10^{-2} \text{ atm}$; B) $5.55 \times 10^{-2} \text{ atm}$; C) $6.33 \times 10^{-2} \text{ atm}$; D) $3.15 \times 10^{-2} \text{ atm}$; E) $8.21 \times 10^{-2} \text{ atm}$

Note: Correct option: A - $4.20 \times 10^{-2} \text{ atm}$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $20.78 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$ '.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '100 g solute / mole solute'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Hydrogen bond < dipole-dipole < induced dipole'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 27.7 g. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 0.0006 volts. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key